

Bounded reciprocal mass of small non-divisor primes of $\binom{2n}{n}$, outside a power-saving exceptional set

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Draft — August 3, 2026

Abstract

Call a prime p *balanced at n* if $p \nmid \binom{2n}{n}$; by Kummer's theorem this holds iff every base- p digit of n is at most $(p-1)/2$. For a constant cutoff B_1 we prove: for every $\varepsilon > 0$ and all large N , all but at most N^ε integers $n \in [N, 2N]$ satisfy

$$\sum_{\substack{3 \leq p \leq B_1 \\ p \nmid \binom{2n}{n}}} \frac{1}{p} \leq \text{LP}(B_1) + \varepsilon, \quad \text{LP}(B_1) := \max \left\{ \sum_{p \in S} \frac{1}{p} : S \subseteq \{3, \dots, B_1\} \text{ prime}, \sum_{p \in S} \beta_p \leq 1 \right\},$$

where $\beta_p = 1 - \log \frac{p+1}{2} / \log p$. One has $\text{LP}(B_1) \leq \text{LP}(\infty) = 0.6941\dots$, against the trivial bound $\sum_{p \leq B_1} 1/p \rightarrow \infty$. The proof identifies joint balance with membership of the mantissa in an intersection of self-similar Cantor sets and applies the higher-rank Furstenberg slicing theorem of Corso–Shmerkin (arXiv:2409.04608, Corollary 1.17), a preprint whose relevant proof chain we have verified independently; the result is conditional on it and on nothing else beyond elementary arithmetic. Empirically the exceptional set appears to be not merely power-saving but finite: the integers balanced at $\{3, 5, 7, 11\}$ simultaneously are $\{1, 3160\}$ up to 10^9 .

1 Statement

Fix once and for all an odd prime bound B_1 and write $\mathcal{P} = \{3 \leq p \leq B_1 : p \text{ prime}\}$. For an odd prime p let

$$D_p = \{0, 1, \dots, \frac{p-1}{2}\}, \quad C_p = \left\{ \sum_{j \geq 1} d_j p^{-j} : d_j \in D_p \right\} \subset [0, 1],$$

the *lower-half-digit Cantor set*, a homogeneous self-similar set with contraction ratio $1/p$, digit set D_p , satisfying the open set condition, with

$$\dim_{\text{H}} C_p = \frac{\log \frac{p+1}{2}}{\log p} = 1 - \beta_p, \quad \beta_p := \frac{\log \frac{2p}{p+1}}{\log p}.$$

Say n is *p -balanced* if every base- p digit of n lies in D_p ; by Kummer's theorem this is equivalent to $p \nmid \binom{2n}{n}$. Let $S(n) = \{p \in \mathcal{P} : n \text{ is } p\text{-balanced}\}$.

Theorem 1. Assume Corollary 1.17 of arXiv:2409.04608 (see Remark ??). For every $\varepsilon > 0$ there are $C_0 = C_0(B_1, \varepsilon)$ and N_0 such that for all $N \geq N_0$,

$$\#\left\{n \in [N, 2N] : \sum_{p \in S(n)} \frac{1}{p} > \text{LP}(B_1)\right\} \leq C_0 N^\varepsilon.$$

Since $\text{LP}(B_1) \leq \text{LP}(\infty) < 0.695$ while $\sum_{p \in \mathcal{P}} 1/p$ is unbounded in B_1 , the theorem separates the balanced mass from its trivial bound by an unbounded margin, for all but a power-saving number of integers.

2 Proof

2.1 Over-budget sets are Cantor intersections

If $\sum_{p \in S(n)} 1/p > \text{LP}(B_1)$ then, by definition of the linear program, $\sum_{p \in S(n)} \beta_p > 1$; discarding elements, $S(n)$ contains a subset S with

$$\sum_{p \in S} \beta_p > 1, \quad \text{i.e.} \quad s := \sum_{p \in S} \dim_{\text{H}} C_p < |S| - 1 =: k - 1. \quad (1)$$

It therefore suffices to bound, for each of the at most $2^{|\mathcal{P}|}$ (a constant) subsets S satisfying (??), the number of $n \in [N, 2N]$ balanced at every $p \in S$.

Fix such an $S = \{p_1, \dots, p_k\}$ and let $n \in [N, 2N]$ be balanced at each p_j . Write $d_j = \lfloor \log 2N / \log p_j \rfloor$, so that $p_j^{d_j} \leq 2N < p_j^{d_j+1}$ and n has at most d_j base- p_j digits. Balance says exactly that

$$\frac{n}{p_j^{d_j}} \in C_p^{(d_j)} \quad (1 \leq j \leq k), \quad (2)$$

where $C_p^{(J)}$ denotes the depth- J truncation of C_p (the union of the $|D_p|^J$ level- J construction intervals).

2.2 The warp: mantissa coordinates

Rescale by the window: put $x = n/(2N) \in [\frac{1}{2}, 1]$. Then (??) reads

$$x \in g_j(C_{p_j}^{(d_j)}), \quad g_j(t) = \frac{p_j^{d_j}}{2N} t, \quad \text{with slopes} \quad \frac{1}{p_j} < \frac{p_j^{d_j}}{2N} \leq 1.$$

This is the decisive normalization: *the affine maps have slopes bounded between $1/B_1$ and 1, independently of N .* (Applied in value space the slopes would be of size N and the uniform constant below would be vacuous.)

2.3 Truncation

The sets in §?? are truncations, not the invariant sets themselves. The bridge is:

Lemma 2 (truncation bridge). *Let $C_1, \dots, C_k$ be the lower-half Cantor sets of distinct odd primes, s as above, $\varepsilon > 0$. There is $C^*(k, \varepsilon, p_1, \dots, p_k)$ such that for all depths $J_j \geq 1$ and all δ with $\max_j p_j^{-J_j} \leq \delta \leq \delta_0$,*

$$N_\delta \left(\bigcap_{j=1}^k g_j(C_j^{(J_j)}) \right) \leq 7 (3\sqrt{k})^{k-1} C^* \delta^{-\max\{s-(k-1), 0\}-\varepsilon},$$

uniformly over affine g_j with slopes in $[1/B_1, 1]$.

Proof. Each point of a depth- J truncation lies within $p_j^{-J_j} \leq \delta$ of C_j (append zero digits; $0 \in D_p$). A 3δ -separated subset of the intersection therefore lifts, coordinatewise, to a δ -separated subset of $(g_1(C_1) \times \dots \times g_k(C_k)) \cap \pi^{-1}(B(0, \delta\sqrt{k}))$, where π is the orthogonal projection killing the diagonal. Covering $B(0, \delta\sqrt{k})$ by $(3\sqrt{k})^{k-1}$ δ -balls and applying the ball-fibre bound underlying Corollary 1.17 of arXiv:2409.04608 (their Lemma 6.1, with the slopes absorbed into the constant via $K = B_1$) bounds each fibre by $C^* \delta^{-\max\{s-(k-1), 0\}-\varepsilon}$; a 3δ -packing/ δ -covering comparison on $\mathbb{R}$ costs the factor 7. $\square$

2.4 Counting and union

Apply Lemma ?? with $\delta = 1/(2N)$ and $J_j = d_j$ (so $p_j^{-d_j} < p_j/(2N) \leq B_1\delta$; replace δ by $B_1\delta$, absorbed in constants). By (??), $\max\{s - (k-1), 0\} = 0$, so the number of δ -cells of $[\frac{1}{2}, 1]$ meeting the intersection is at most $C_1(B_1, \varepsilon) N^\varepsilon$. Each δ -cell has length $1/(2N)$ and thus contains at most one value of $x = n/(2N)$ with n integral. Hence at most $C_1 N^\varepsilon$ integers $n \in [N, 2N]$ are balanced at all of S . Summing over the constantly many over-budget S gives the theorem with $C_0 = 2^{\pi(B_1)} \max_S C_1$. $\square$

3 Remarks

Remark 3 (dependency). *The input is Corollary 1.17 of E. Corso and P. Shmerkin, Dynamical self-similarity, L^q -dimensions and Furstenberg slicing in $\mathbb{R}^d$, arXiv:2409.04608 (2024), a preprint at the time of writing. We have verified the proof chain we consume (Cor. 1.17 $\Leftarrow$ Thm. 1.15 $\Leftarrow$ Thm. 5.4 $\Leftarrow$ their inverse-theorem engine) for the special case used here, in the course of which two repairable defects were identified (a misprinted hypothesis (P1) of their §5, whose intended form is confirmed by their Remark 1.16, and an omitted almost-everywhere injectivity verification in the last step of Theorem 5.4, which we supply for our projection); details in the project's verification ledger. No other non-elementary input occurs.*

Remark 4 (what is and is not proved). *The theorem bounds the balanced mass of primes up to a constant B_1 , for all n outside a power-saving exceptional set. It does not bound the mass over the growing range $p \leq (\log n)^{O(1)}$ (which requires a base-uniform, effective form of the slicing theorem), and it does not empty the exceptional set (a rigidity question). Empirically the exceptional sets appear finite and tiny: the integers balanced simultaneously at $\{3, 5, 7, 11\}$ — the smallest over-budget set, $\sum \beta = 1.227$ — are $\{1, 3160\}$ up to 10^9 , and at $\{3, 5, 7, 13\}$ are $\{1, 756, 757, 3250\}$ up to 10^8 .*

Remark 5 (context). $\text{LP}(\infty) = 0.6941\dots$ matches, to the accuracy of measurement, the empirical ceiling of the balanced mass located by exhaustive search in earlier stages of this programme ($\approx 0.69\text{--}0.83$ across normalizations), suggesting the linear program is the true extremal mechanism. The theorem is a step in a programme to prove $\sum_{p \mid \binom{2n}{n}} 1/p = O(1)$ for all n (the bounded form of Erdős problem 377); the asymptotic form is treated in a companion paper.